

# Machine Learning–Based Excess Path Loss Modeling for S-Band Terrestrial Radio Propagation

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**Abstract**—Accurate radio propagation modeling is essential for the design and planning of modern terrestrial and satellite-assisted communication systems operating in the S-band. Classical empirical models such as the COST-231 Hata model often fail to account for environment-dependent excess losses caused by atmospheric and local propagation effects, leading to significant prediction errors. In this work, a data-driven framework is proposed to model excess path loss beyond conventional propagation models using supervised machine learning. A synthetic propagation dataset is generated by combining free-space path loss, COST-231 predictions, and additional attenuation due to rain and atmospheric parameters. Excess loss is defined as the deviation between the total simulated path loss and the COST-231 baseline. Multiple regression models are evaluated, including linear regression and random forest regression. Experimental results demonstrate that the random forest model accurately captures the nonlinear behavior of excess loss, achieving a root mean square error of 0.035 dB, significantly outperforming traditional empirical approaches. The proposed framework highlights the potential of machine learning techniques to enhance classical propagation models for improved radio link planning.

**Index Terms**—Radio propagation, S-band communication, COST-231 model, excess path loss, machine learning, random forest regression

## I. INTRODUCTION

Reliable radio propagation modeling is a fundamental requirement for the design and deployment of modern wireless communication systems. Accurate estimation of path loss directly impacts link budget analysis, coverage planning, and quality-of-service guarantees. In frequency bands such as the S-band, commonly used for satellite communication, telemetry, and emerging terrestrial applications, propagation characteristics are influenced by both free-space attenuation and environmental factors such as atmospheric absorption, rain attenuation, and terrain-induced losses.

Traditional empirical propagation models, including the COST-231 Hata model, provide simplified path loss estimation based on large-scale measurements. While these models are computationally efficient and widely adopted, they often fail to capture excess losses arising from localized environmental variations. As a result, significant discrepancies can occur between predicted and actual received signal levels, particularly in dynamic or non-ideal propagation conditions.

Recent advancements in machine learning have enabled data-driven modeling approaches capable of capturing nonlinear relationships in complex systems. By learning from

propagation data, machine learning models can complement classical empirical models and improve prediction accuracy. This work explores the application of supervised learning techniques to estimate excess path loss beyond the COST-231 baseline for S-band terrestrial propagation.

## II. RELATED WORK

Radio propagation modeling has been extensively studied using empirical, deterministic, and stochastic approaches. Empirical models such as the Hata and COST-231 models are derived from large-scale measurement campaigns and are widely used due to their simplicity. However, their applicability is limited to specific frequency ranges and environments, and they do not explicitly model excess attenuation effects.

In recent years, machine learning techniques have been applied to propagation modeling to overcome these limitations. Regression-based models, artificial neural networks, and ensemble learning methods have been explored to predict path loss using environmental and system parameters. Studies have shown that machine learning models can outperform classical models when sufficient training data is available.

Despite these advances, limited work has focused on explicitly modeling excess path loss as a separate quantity relative to classical empirical predictions. This work addresses this gap by defining excess loss with respect to the COST-231 model and employing supervised learning to predict it using physically meaningful parameters.

## III. SYSTEM MODEL AND DATASET GENERATION

The proposed system integrates classical propagation modeling with machine learning-based excess loss estimation. A synthetic dataset is generated to represent S-band terrestrial propagation under varying environmental conditions. The total path loss is computed as the sum of free-space path loss, COST-231 predicted loss, and additional attenuation components.

Fig. 1 illustrates the overall system architecture, including dataset generation, excess loss computation, and machine learning model training.

The dataset includes parameters such as link distance, rain rate, humidity, temperature, and corresponding path loss components. Excess loss is defined as the difference between the total simulated path loss and the COST-231 predicted loss.

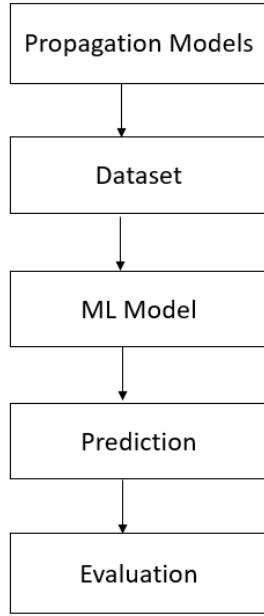


Fig. 1. System overview of the proposed excess path loss modeling framework.

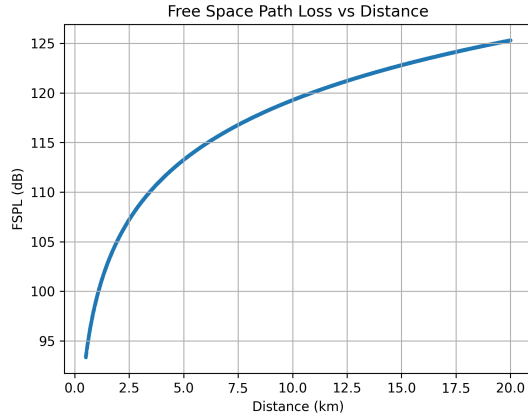


Fig. 2. Free-space path loss versus distance for S-band frequency.

#### IV. PROPAGATION MODELS AND EXCESS LOSS DEFINITION

Free-space path loss (FSPL) represents the fundamental attenuation of radio waves in an unobstructed environment and is calculated as a function of frequency and distance. The COST-231 Hata model extends the original Hata model to higher frequencies and urban environments, providing an empirical estimate of large-scale path loss.

Fig. 2 illustrates the variation of free-space path loss with distance, while Fig. 3 compares COST-231 predictions with the total path loss.

Excess path loss is defined as the residual component that accounts for additional attenuation effects not captured by the

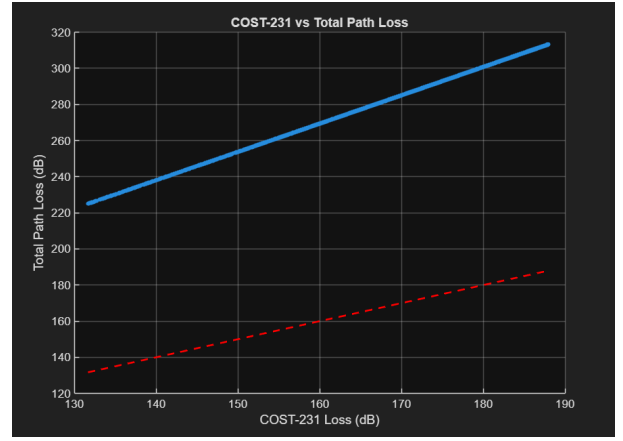


Fig. 3. Comparison of COST-231 predicted loss and total path loss.

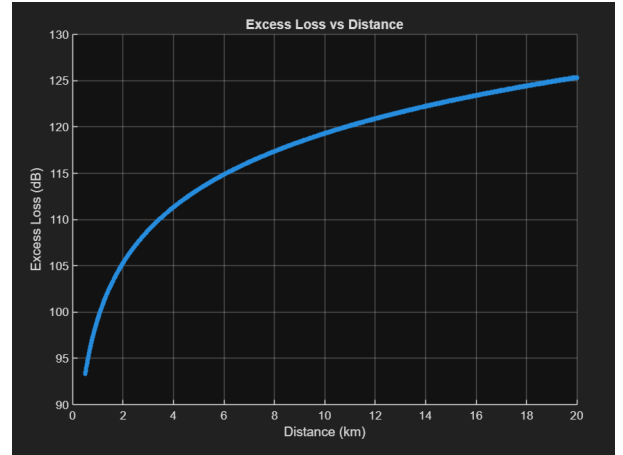


Fig. 4. Excess path loss variation with distance.

COST-231 model.

#### V. MACHINE LEARNING FRAMEWORK

Supervised regression models are employed to predict excess path loss using environmental and link parameters as inputs. Linear regression serves as a baseline model, while random forest regression is used to capture nonlinear relationships.

Fig. 4 shows the variation of excess loss with distance, highlighting nonlinear behavior.

The dataset is divided into training and testing subsets, and model performance is evaluated using standard regression metrics.

#### VI. RESULTS AND DISCUSSION

The random forest model demonstrates superior prediction accuracy compared to linear regression. Fig. 5 illustrates the predicted versus true excess loss values, showing close alignment along the ideal prediction line.

Residual error analysis confirms that the prediction errors are centered around zero with low variance, indicating robust model performance.

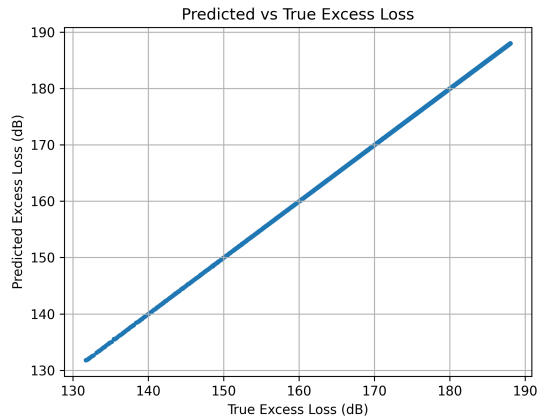


Fig. 5. Predicted versus true excess path loss using the random forest model.

## VII. CONCLUSION AND FUTURE WORK

This work presented a machine learning-based framework for modeling excess path loss in S-band terrestrial radio propagation. By combining classical empirical models with data-driven regression techniques, improved prediction accuracy was achieved. The results demonstrate that random forest regression effectively captures nonlinear excess loss behavior beyond the COST-231 baseline.

Future work will focus on incorporating real-world measurement data, extending the framework to other frequency bands, and exploring deep learning architectures for enhanced generalization.

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